

TECHNICAL PROPOSAL AND SOFTWARE REQUIREMENTS AND DESIGN DOCUMENT

Shaba Village Estate Platform

Unit sales and instalments, utility and service charge billing, M-Pesa collections, service provider management, gate security and owner self-service for Shaba Village, Syokimau.

PREPARED FOR

Shaba Village
Estate management

PREPARED BY

Codevertex Africa Limited
www.codevertexafrica.com

DATE

6 October 2026

DELIVERY PERIOD

14 working weeks

INVESTMENT

KES 350,000 build
From KES 10,000 per month

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Document control

Version	Date	Author	Description
1.0	6 October 2026	Codevertex Africa Limited	Issued for client review

Role	Party	Responsibility
Product owner	Shaba Village estate management	Scope approval, business rules, by-law and tariff decisions, data supply, legal sign-off
Finance lead	Shaba Village	Charge rates, opening balances, bank and paybill set-up, approval limits
Delivery lead	Codevertex	Planning, scope control, reporting, risk management
Technical lead	Codevertex	Architecture, code quality, security, releases

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Executive summary

Codevertex Africa Limited proposes to design, build and launch the Shaba Village Estate Platform. The platform runs the estate's two money streams and its daily operations in one place. The first stream is the sale of homes and apartments, paid outright or by deposit and instalments. The second is the monthly billing of water, service charge, garbage collection, security and other estate charges to every owner.

Owners see their purchase progress, bills, meter readings and receipts on their phone and pay by M-Pesa against their unit number. Payments are matched and receipted automatically. The management team works from one console covering units, billing, arrears, sales, maintenance, service providers, gate security and reports. Security, cleaning and garbage agencies work through the same platform, and their invoices are checked against the service they actually delivered before payment.

The platform is built on Codevertex's production services for sign-in, accounting, M-Pesa collections, payroll and messaging. Only the estate-specific service and its screens are new, which keeps the build price and the delivery time low.

Key outcomes

- Every unit carries one balance per fund, visible to the owner at any time, with statements on demand.
- Paybill payments matched to the right unit and receipted within a minute, with no manual reconciliation for correctly referenced payments.
- Monthly bills, including metered water, issued for all units in one run on the 1st, with reading photos as evidence.
- Instalment schedules, reminders and purchase statements for every buyer, from reservation to handover and title.
- Service provider contracts, schedules, guard posts and patrols tracked, and vendor invoices adjusted for missed service before payment.
- Visitor passes and gate logs that protect residents while collecting only the data the law allows.
- Estate staff, including casual workers, paid through the ERP payroll with statutory deductions.
- Sales money and estate service money kept in separate accounts and ledgers, ready for handover to the owners' management corporation.

Engagement summary

14 weeks Delivery period, launch in February 2027	KES 350,000 Fixed build price, inclusive of taxes	KES 20,000 Recommended monthly fee: subscription, hosting, support	KES 10,000 Entry monthly tier, up to 150 units
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SECTION 1

Introduction

1.1 Purpose

This document specifies the requirements, design, delivery plan and commercial terms for the Shaba Village Estate Platform. It is the basis for development, testing and acceptance, and replaces the earlier generic property management requirements document.

1.2 About Shaba Village

Shaba Village is a residential estate in Syokimau, off Mombasa Road, in the Mavoko area of Machakos County. It comprises homes and apartments, including two and three bedroom units. Units are sold outright or on a deposit and instalment basis. Estate management supplies water from the Mavoko Water and Sewerage Company connection and two boreholes, stored in an underground reservoir and unit tanks, and runs shared services: perimeter security with electric fencing, 24-hour guarding, landscaping, cleaning of common areas, garbage collection, sewer and parking. These services are billed to owners every month.

1.3 Product scope

Area	Included
Estate register	Blocks, houses and apartments, unit types and entitlements, owners and joint owners, occupants and tenants, household members, vehicles, domestic staff
Unit sales	Price lists, reservations, sale agreements, deposits, instalment schedules, purchase statements, late payment handling, handover, defects and title progress
Billing and collections	Service charge, water, garbage, security levy, sinking fund, parking and special levies; meter readings; monthly billing runs; M-Pesa paybill and prompt payments; allocation; receipts; statements; arrears and payment plans; clearance certificates
Service providers	Vendor onboarding and licence tracking, contracts and SLAs, garbage and cleaning schedules, security posts, rosters and patrols, service evidence, vendor portal, invoice checking, approvals and payment
Maintenance	Resident requests, work orders, preventive schedules for pumps, tanks and fences, quotes, completion evidence, recharges to units, defects liability claims
Security and access	Gate tablet application, visitor passes, walk-in approval, deliveries, contractor access, vehicle log, incident reports, digital occurrence book
Estate staff	Caretakers, plumbers, gardeners and casual workers in the ERP: records, attendance, payroll, casual payments, statutory deductions
Communication	Notices by block or estate, water interruption and emergency alerts, AGM notices, document library
Finance and reporting	Separate ledgers for sales and estate funds, service charge budget against actual, income and expenditure for the AGM, collections, arrears, water balance, vendor performance
Compliance	Consent records, privacy notices, data requests, retention, access logging, legal policy drafts, WCAG 2.2 AA accessibility

Excluded: native iOS and Android store applications (the web application installs on the home screen instead), smart meter and CCTV hardware, conveyancing and legal advice, rent collection on behalf of owners who let their units, marketing of units to the public, and land or property listings for third parties.

1.4 Definitions

Term	Definition
Unit	A house or apartment with its own code, for example B07 (block B, unit 7)
Owner	The registered or contracted purchaser of a unit; joint owners share one unit account
Occupant	A person living in the unit: the owner, a household member or a tenant of the owner
Unit entitlement	The share of common costs carried by a unit, set by unit type or floor area
Service charge	The monthly contribution to the cost of running the estate
Sinking fund	A reserve for major repairs and replacement of shared assets
Estate fund	Money collected for estate services, held apart from sale proceeds
Billing run	The monthly process that issues one invoice per unit for all estate charges
Water balance	Water supplied by Mavoko and the boreholes compared with water billed to units
SLA	Service level agreement: measurable targets in a provider contract
Work order	A recorded job for repair or maintenance
Management corporation	The owners' body corporate formed under the Sectional Properties Act, 2020
Paybill, STK push	M-Pesa business number; payment prompt sent to the payer's phone
ODPC, DPIA	Office of the Data Protection Commissioner; data protection impact assessment
eTIMS, WHT	KRA electronic tax invoice system; withholding tax
RPO, RTO, UAT	Recovery point and recovery time objectives; user acceptance testing

1.5 References

- Property Management and Real Estate Services Platform requirements document, version 1.0, October 2026 (superseded by this document)
- Codevertex platform documentation: microservice architecture, cross-service data ownership, treasury invoicing and payments, M-Pesa integration, approvals, ERP payroll, notifications
- Sectional Properties Act, 2020; Data Protection Act, 2019 and its Regulations, 2021; ODPC draft Guidance Note for the Private Security Sector, 2025
- Mavoko Water and Sewerage Company approved tariff, 2023/24 to 2025/26
- Web Content Accessibility Guidelines 2.2; OWASP Application Security Verification Standard 4

SECTION 2

Overall description

2.1 Product perspective

Shaba Village operates as a tenant of the Codevertex platform. Two new components, estates-ui and estates-api, provide the estate product. Sign-in, invoicing, M-Pesa collections, the ledger, approvals, payroll and messaging are provided by existing platform services, described in [Solution architecture](#). The estate service holds references to their records and never keeps a second copy.

2.2 User classes

Role	Description	Main permissions	Sign-in
Owner	Purchaser of a unit, resident or living elsewhere, including diaspora owners	Statements, payments, purchase progress, meter readings, requests, visitor passes, household, documents, privacy	Phone and OTP
Occupant	Household member or tenant nominated by the owner	Visitor passes, requests, notices; bills only if the owner assigns them	Phone and OTP
Buyer	Purchaser before handover	Purchase statement, instalment schedule, payments, documents, handover snag list	Phone and OTP
Estate manager	Runs daily operations	All operations, work orders, vendors, notices, reports	SSO with MFA
Finance officer	Billing and collections	Charges, billing runs, allocations, adjustments within limits, vendor bills, reports	SSO with MFA; OTP per approval
Sales officer	Unit sales	Price lists, reservations, agreements, instalment plans, buyer documents	SSO with MFA
Caretaker	On-site staff lead	Meter readings, work orders, service visit checks, notices	SSO, phone
Guard	Agency guard on duty	Gate tablet only: passes, walk-ins, deliveries, patrols, incidents	Gate PIN on a registered tablet
Vendor supervisor	Agency or contractor representative	Own contracts, schedules, rosters, work orders, invoices, documents	Phone and OTP
Administrator	Shaba Village directors and delegates	All permissions, staff roles, settings, approvals above limit	SSO with MFA

2.3 Operating environment

- Owners, occupants and vendors: any modern mobile or desktop browser; installable as a progressive web application
- Gate: an Android tablet at each gate in kiosk mode, with offline operation during network loss
- Hosting: Codevertex Kubernetes platform with PostgreSQL, Redis, NATS JetStream and object storage, behind Cloudflare
- Payments: Safaricom M-Pesa through Shaba Village's own paybills (Daraja API); bank transfers recorded against unit accounts
- Messaging: Africa's Talking (SMS), Meta WhatsApp Cloud API, Codevertex mail server (email)

2.4 Constraints

- Money is collected only into Shaba Village's own paybills and bank accounts. Codevertex never receives estate or buyer funds.
- The estate service does not move money. Invoices, payments and payouts are executed by the treasury service.
- Utility supply is never cut off automatically for arrears. Kenyan courts and tribunals have held that disconnection without due process is unlawful.
- Electricity is supplied by Kenya Power prepaid meters per unit and is not resold by the estate.
- Conveyancing, title registration and legal notices are performed by advocates; the platform records their progress.

2.5 Assumptions and dependencies

- Shaba Village supplies the unit register, owner list, sale agreements and opening balances by week 4, in the import template provided.
- Shaba Village holds, or opens, separate paybills and bank accounts for sale collections and for the estate fund, and grants Daraja API access by week 3.
- Shaba Village confirms the charge schedule, water tariff, penalty rules and allocation order from its by-laws and sale agreements by week 5.
- Service provider contracts and licences are made available for loading by week 8.
- Shaba Village appoints a product owner able to decide within 2 working days and documents within 3.
- Shaba Village registers with the ODPC as a data controller and engages an advocate to approve the legal pack.
- Shaba Village provides one Android tablet per gate, with a data connection.
- Messaging, payment provider charges and the domain name are client-borne costs, as set out in [Commercial terms](#).

SECTION 3

Estate operating model

This chapter sets out who pays what, into which account, and how Shaba Village's two money streams stay apart.

3.1 Charges

Charge	Billed to	Basis	Fund
Unit price	Buyer	Outright, or deposit plus instalments per the sale agreement	Sales
Reservation fee	Prospective buyer	Fixed, credited to the deposit when the agreement is signed	Sales
Service charge	Owner	Fixed per unit type, or annual budget divided by unit entitlement	Estate
Water	Owner, or the occupant if the owner assigns it	Metered: consumption times the estate tariff, plus any fixed meter charge	Estate
Garbage collection	Owner or occupant	Fixed per unit, or included in service charge	Estate
Security levy	Owner	Fixed per unit, or included in service charge	Estate
Sinking fund	Owner	Fixed or percentage of service charge, kept in its own ledger account	Estate (reserve)
Extra parking	Owner or occupant	Per additional bay	Estate
Special levy	Owner	One-off amount approved at a general meeting, by entitlement	Estate
Recharges	Owner	Cost of in-unit repairs or damage, with the work order as evidence	Estate
Late payment charge	Owner or buyer	Only where the by-laws or sale agreement provide; capped and never compounded	As the debt

Every charge type, rate and effective date is configurable by finance. A rate change applies from its effective date and never alters invoices already issued.

3.2 Worked example: monthly bill

Unit B07, three bedroom, October 2026	KES
Service charge, three bedroom rate	4,500
Water: reading 1,284 less 1,275 = 9 m3 at KES 150	1,350
Garbage collection	300
Sinking fund	300
Total due by 10 November 2026	6,450

Rates are illustrative and are set by Shaba Village. For comparison, the Mavoko Water and Sewerage Company approved domestic tariff ranges from KES 120 per m3 for the first 6 m3 to KES 170 per m3 above 300 m3.

3.3 Who receives the bill

The owner is liable for all estate charges by default. Where an owner lets the unit, the owner may assign water, garbage and parking to the tenant, who then receives those bills and pays with the same unit account number. Service charge, sinking fund and special levies remain with the owner. If an assigned tenant leaves with an unpaid balance, the balance reverts to the owner after 30 days, as the owner terms state.

3.4 Fund segregation

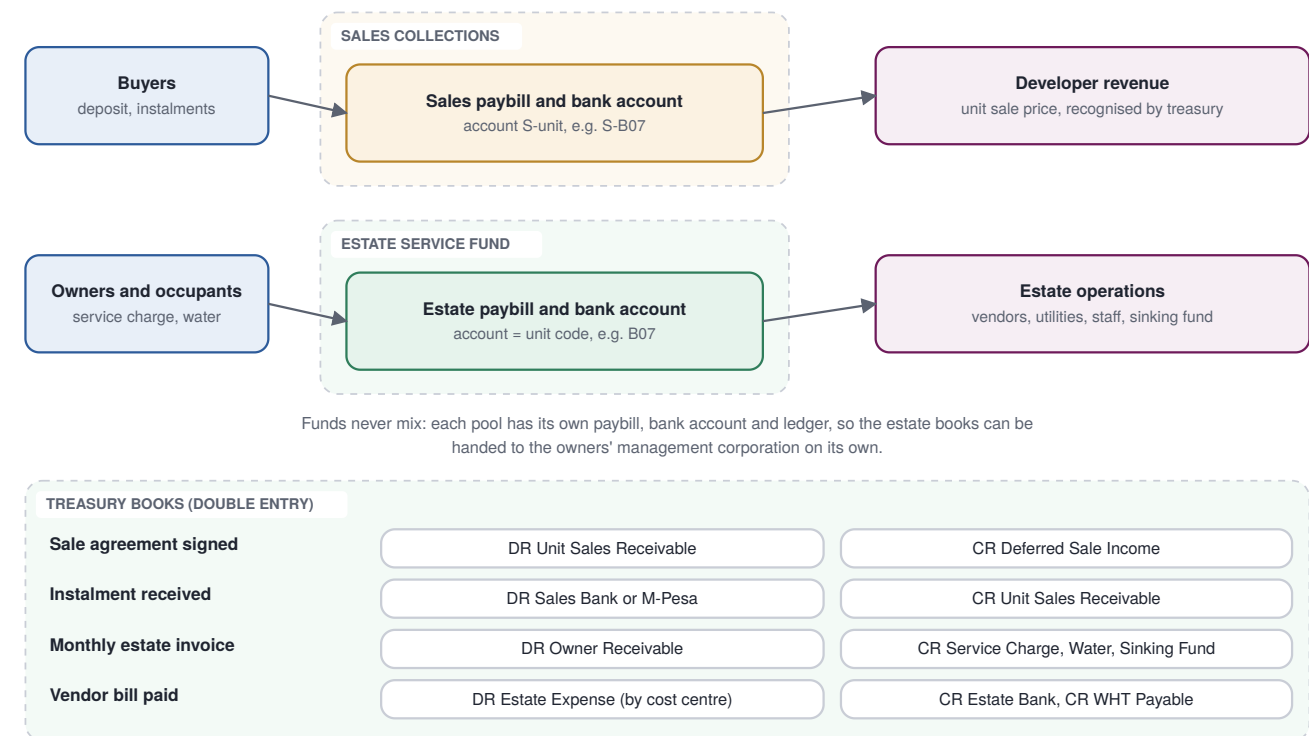


Figure 1. Fund segregation and accounting

Sale proceeds belong to the developer. Estate charges are collected to run the estate on behalf of all owners. Mixing the two is the most common cause of service charge disputes in Kenyan estates. The platform keeps them apart by design: separate paybills and bank accounts, separate cost centres and ledger accounts in treasury, and separate statements. A buyer who is also an owner sees both balances, but a payment to one account can never settle the other.

3.5 Payment references

Payment	Paybill	Account number
Estate charges	Estate paybill	Unit code, for example B07
Unit purchase	Sales paybill	S plus unit code, for example S-B07

Owners may also pay by M-Pesa prompt from the portal, which needs no reference, or by bank transfer with the unit code as reference. Payments that cannot be matched are placed in a suspense queue for finance, with the payer's name and number to assist.

3.6 Transition to the management corporation

Under the Sectional Properties Act, 2020, common property in a sectionally registered estate is owned and managed by a management corporation of the unit owners. The estate fund books, unit accounts, sinking fund, vendor contracts and records are kept so that they can be handed over complete when the corporation is formed. Shaba Village can then continue as managing agent for the corporation on the same platform, or hand over administrative control entirely.

SECTION 4

Functional requirements

Priority: **M** must have, **S** should have, **C** could have. All M and S requirements are in the build price.

ID	Requirement	Pri.
ESTATE REGISTER AND ACCOUNTS		
FR-01	Record blocks, houses and apartments with code, type, bedrooms, floor area, parking bays, meter serials and unit entitlement.	M
FR-02	Record owners, joint owners and ownership dates per unit. A change of ownership keeps the history and closes the old owner's account on the transfer date.	M
FR-03	Record occupants (owner, household member, tenant) with dates. Owners add and remove their own occupants, vehicles and domestic staff.	M
FR-04	Import units, owners, occupants, sale contracts and opening balances from a spreadsheet template, with validation and a dry run before commit.	M
FR-05	Owners, buyers, occupants and vendor supervisors sign in with a mobile number and a 6-digit one-time password. Staff use single sign-on with MFA.	M
FR-06	Owners accept the terms and privacy notice on first sign-in; version and time are recorded.	M
UNIT SALES AND INSTALMENTS		
FR-07	Maintain price lists by unit type and phase with effective dates. Unit availability shown as available, reserved, under agreement, fully paid, handed over and titled.	M
FR-08	Reserve a unit for a named buyer for a set period on payment of a reservation fee. Expired reservations release the unit automatically.	M
FR-09	Record a sale contract: buyer KYC, price, discounts, deposit, payment terms, agreement date, advocates, and the signed agreement document.	M
FR-10	Generate an instalment schedule (monthly, quarterly or milestone based) or record an outright purchase. Schedules can be restructured with approval; the original is kept.	M
FR-11	Buyers pay instalments by sales paybill or M-Pesa prompt. Payments settle the oldest due instalment first; overpayments reduce the next instalment.	M
FR-12	Reminders before each due date and after a missed one. Late charges apply only as the agreement provides. Defaults beyond the grace period are escalated to the sales officer.	M
FR-13	Purchase statement in PDF and spreadsheet: price, payments with M-Pesa receipts, balance and next due date.	M
FR-14	Handover checklist with meter readings, keys, snag list and owner signature; estate billing starts on the handover date, prorated for the first month.	M
FR-15	Track title stages (sectional plan, lease or title preparation, stamp duty, registration, release) with dates and documents.	S
FR-16	Defects raised during the defects liability period are logged as work orders against the contractor.	S
BILLING AND UTILITIES		

FR-17	Configure charge types, rates, bases (fixed, per type, by entitlement, metered), effective dates, fund and bill-to rules.	M
FR-18	Caretakers record monthly meter readings on a phone with a photo per meter. Readings lower than the last, zero, or above three times the 3-month average are flagged.	M
FR-19	Missing readings are estimated from the 3-month average, marked as estimated on the bill, and corrected on the next actual reading.	M
FR-20	Record bulk readings for the Mavoko connection and each borehole; produce a water balance with loss percentage and an alert above the set threshold.	M
FR-21	Run the monthly billing for all units: one invoice per unit and fund, with line items, meter reading, photo link and due date. Reruns never duplicate an invoice.	M
FR-22	Issue one-off invoices for special levies, recharges and parking, and credit notes for corrections, under approval above a limit.	M
FR-23	Owners view each bill with the meter photo and may query a line within 7 days; queries enter the finance queue.	S

PAYMENTS AND COLLECTIONS

FR-24	Match paybill payments by account number to the unit account, allocate by the configured order (oldest first by default), and receipt by SMS or WhatsApp.	M
FR-25	Owners pay from the portal by M-Pesa prompt; finance records bank transfers and cheques with references.	M
FR-26	Unmatched payments go to a suspense queue; assigning one to a unit allocates and receipts it.	M
FR-27	Overpayments are held as credit and applied to the next invoice.	M
FR-28	Escalating reminders at configurable days after the due date; a demand letter generated at a set age for review and sending.	M
FR-29	Arrears payment plans agreed with an owner, with their own schedule and reminders.	S
FR-30	Clearance certificate for a unit resale, issued only when all estate balances are nil.	S
FR-31	Owner statement per fund, on demand, in PDF and spreadsheet.	M

SERVICE PROVIDERS

FR-32	Register vendors with KRA PIN, contacts, payment details and compliance documents (NEMA waste licence, PSRA licence, county permit, NCA registration, insurance), each with an expiry date and alerts 30 days ahead.	M
FR-33	Record vendor contracts: scope, fee basis (monthly, per visit, per job), start and end, SLAs, service credits and renewal date.	M
FR-34	Generate recurring service visits from contract schedules: garbage collection days, cleaning rosters by block, landscaping, pest control, tank cleaning, fence and pump servicing.	M
FR-35	Record visit completion with time and photos by the vendor supervisor or caretaker. A visit not completed in its window is marked missed. Residents can report a missed collection.	M
FR-36	Security agency: define posts and shifts, receive the agency's roster, sign guards on and off at the gate tablet, and report post coverage against the contract.	M
FR-37	Patrol checkpoints with QR tags; scheduled rounds; scans recorded with time; missed rounds reported.	S

FR-38	Register agency personnel deployed at the estate (name, agency, role, phone), issue access badges and deactivate them on removal.	M
FR-39	Vendors submit invoices through the vendor portal with their eTIMS invoice number. The platform shows the contract fee, delivered service, missed service and the resulting service credit before approval.	M
FR-40	Approved vendor bills are paid by bank transfer or M-Pesa from the estate account, with withholding tax where applicable, and a remittance advice sent to the vendor.	M
FR-41	Vendor scorecard: on-time visits, missed visits, post coverage, incidents and resident complaints by month.	S
MAINTENANCE		
FR-42	Residents raise maintenance requests with category, description and photos, and follow progress.	M
FR-43	Work orders carry priority with SLA targets, are assigned to estate staff or a vendor, and record time on site, parts, cost and completion photos.	M
FR-44	Jobs above a set value need a quote and approval before work starts.	M
FR-45	Residents confirm completion or reopen within 7 days. In-unit work can be recharged to the unit.	M
FR-46	Preventive maintenance schedules for shared assets (pumps, boreholes, tanks, reservoir, electric fence, gate, drainage, fire equipment) generate work orders automatically.	M
SECURITY AND ACCESS		
FR-47	Residents create visitor passes (single visit or recurring) with a time window; the visitor receives a QR code and a 6-digit code by SMS or WhatsApp.	M
FR-48	Guards verify passes at the gate, record walk-in visitors and request host approval in the app or by SMS. Entry and exit are logged with plate number where applicable.	M
FR-49	Deliveries, domestic staff and contractors on an open work order have their own pass types.	M
FR-50	The gate tablet works offline for at least 8 hours, using cached passes, and synchronises on reconnection.	M
FR-51	Guards and staff record incidents with category, time, location and photos in a digital occurrence book; serious incidents alert the estate manager at once.	M
FR-52	Estate management may restrict a person from entry with a recorded reason and review date.	S
ESTATE STAFF		
FR-53	Estate staff and casual workers are managed in the ERP: records, contracts, leave, attendance, payroll with PAYE, NSSF, SHIF and the housing levy, and casual daily payments by M-Pesa.	M
FR-54	Work orders and service checks can be assigned to ERP staff; staff sign-on at the gate is shared with ERP attendance.	S
COMMUNICATION AND DOCUMENTS		
FR-55	Send notices to the whole estate, a block, owners only or occupants, by SMS, WhatsApp, email and in-app, with delivery status.	M
FR-56	Water interruption and emergency alerts with priority delivery.	M
FR-57	Document library: by-laws, house rules, AGM notices and minutes, budgets, audited accounts, insurance certificates, with access by role.	M
FR-58	Non-binding owner polls with one response per unit.	C

FINANCE, REPORTS AND ADMINISTRATION

FR-59	Annual service charge budget by cost line; budget against actual every month; income and expenditure statement for the AGM.	M
FR-60	Reports and dashboards as set out in Reporting and analytics , exportable to PDF, CSV and Excel.	M
FR-61	Settings: charges, tariffs, penalties, allocation order, reminder days, approval limits, SLA targets, retention periods.	M
FR-62	Data subject requests (access, correction, deletion) with deadline tracking.	M
FR-63	Audit log of every change to charges, invoices, allocations, sale contracts, approvals, access rights and settings.	M

SECTION 5

Non-functional requirements

ID	Attribute	Requirement
NFR-01	Mobile performance	Owner portal initial transfer under 200 KB excluding images. Largest Contentful Paint under 2.5 seconds on a mid-range Android phone over 3G. Photos served as resized WebP.
NFR-02	Response time	95th percentile under 300 ms for reads and 600 ms for writes, excluding M-Pesa latency. Gate pass verification under 1 second, online or offline.
NFR-03	Capacity and growth	Designed for 2,000 units, 100,000 gate events and 5,000 payments a month, scalable tenfold without redesign: keyset pagination on all lists, monthly partitions for gate events, reports from daily aggregates, read replica for analytics.
NFR-04	Billing performance	A billing run for 1,000 units completes within 2 minutes and is safe to retry.
NFR-05	Availability	99.5% monthly for the portal, gate service and payment matching. Two replicas per service; zero-downtime deployments.
NFR-06	Financial integrity	Each M-Pesa receipt recorded once; each invoice issued once per unit and period; balanced double-entry ledger; daily reconciliation of paybill statements against payments.
NFR-07	Security	OWASP ASVS Level 2, TLS throughout, encryption of identity numbers and KRA PINs at field level, MFA for staff, audit trail on financial and configuration changes.
NFR-08	Privacy	Data minimisation, consent records, retention schedule enforced by automated purge, documented hosting location and transfer safeguards.
NFR-09	Accessibility	WCAG 2.2 Level AA.
NFR-10	Localisation	English at launch; text externalised for Kiswahili. Dates and amounts in Kenyan format.
NFR-11	Recovery	Daily encrypted offsite backups retained as 7 daily and 4 weekly copies; quarterly restore drills. RPO 24 hours, RTO 4 hours.
NFR-12	Observability	Structured logs, tracing, alerts on failed billing runs, unmatched payment build-up, offline gate devices, error rates and latency.
NFR-13	Maintainability	Codevertex platform stack and conventions, automated tests, documented APIs, versioned database migrations.

SECTION 6

Solution architecture

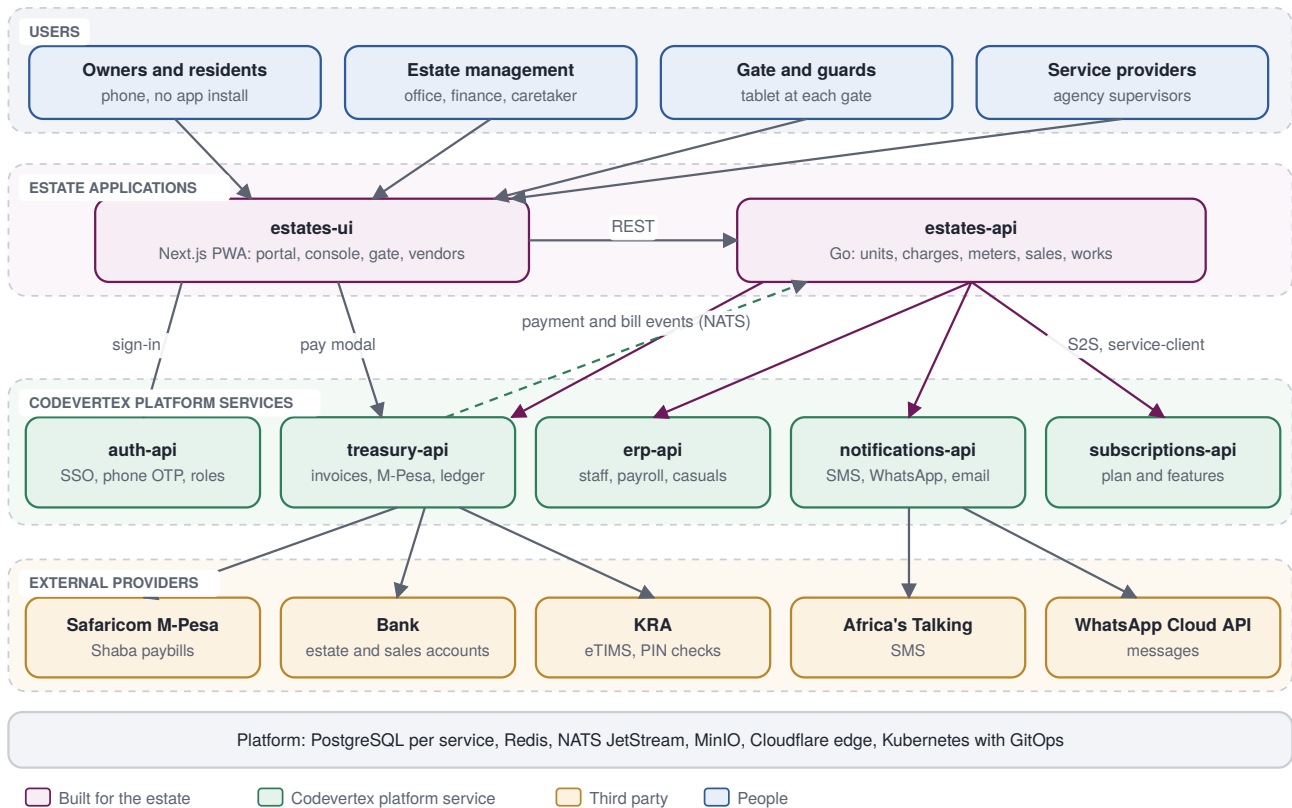


Figure 2. Solution architecture

6.1 Components

Component	Status	Function
estates-ui	New	Next.js progressive web application with four surfaces: owner portal, management console, gate tablet and vendor portal. Uses the Codevertex shared UI library for sign-in, the payment modal, data tables with export, document preview, legal links and cookie consent.
estates-api	New	Go service for units, ownership, occupancy, charge rules, meters, billing runs, sale contracts, work orders, vendors' contracts and visits, agency personnel, posts and patrols, passes, gate events, incidents, notices and statistics. Does not move money.
auth-api	Existing, extended	Tenant, staff single sign-on with MFA, roles, sessions, terms acceptance. Extended with mobile one-time password sign-in for owners, occupants and vendor supervisors.
treasury-api	Existing, extended	Customers and balances, invoices and credit notes, recurring billing, payment intents, M-Pesa prompt and paybill (C2B), allocation, dunning reminders, statements, vendor bills, approvals, payouts, withholding tax, eTIMS, budgets and the general ledger. Extended with paybill matching by unit account number and instalment schedules linked to sale contracts.
treasury-ui	Existing	Finance console for the ledger, bank reconciliation, vendor bills, approvals and financial statements.

erp-api, erp-ui	Existing	Estate staff records, contracts, attendance, leave, payroll with statutory deductions, casual worker payments and asset register. Payroll posts to treasury.
notifications-api	Existing	SMS, WhatsApp, email and push delivery with templates and delivery status. Estate templates added.
subscriptions-api	Existing	Shaba Village's plan, feature entitlements and limits.
Shared libraries	Existing	Authentication client, service client with retries and circuit breakers, event outbox, cache with job leases, rate limiting, pagination, HTTP middleware, shared UI library.

6.2 Architecture principles

- **Single data owner.** Each record has one home. estates-api stores the treasury invoice, account and plan IDs, the ERP employee ID and the auth user ID, and reads balances from treasury rather than keeping its own.
- **Single payment path.** Only treasury-api talks to M-Pesa and the banks. estates-api asks treasury to invoice and to pay.
- **Event driven.** Treasury publishes payment, invoice and payout events through a transactional outbox to NATS. estates-api consumes them idempotently by event ID.
- **Replica safe.** Billing runs, reminders, SLA timers and purges run once per period across all replicas using shared leases.
- **Built for growth.** Indexed tenant and estate keys on every table, keyset pagination, time-partitioned event tables, aggregates for reports and a read replica for heavy queries.
- **Standard stack.** Go with Ent and Atlas migrations, Next.js, Helm and ArgoCD.

6.3 What already exists

Most of what an estate needs financially is already in production on the platform. The table shows where the estate build reuses it.

Need	Existing capability used
Monthly invoices with numbering, tax and due dates	treasury invoicing and recurring schedules
M-Pesa prompts and paybill confirmations, recorded once per receipt	treasury M-Pesa gateway and C2B inbox
Escalating overdue reminders with a pay link	treasury dunning worker
Owner and vendor balances, ageing and statements	treasury receivables and payables
Vendor bills, approval with OTP, payment by bank or M-Pesa	treasury vendor bills, approvals engine and payout dispatcher
Withholding tax and eTIMS	treasury tax module and KRA integration
Instalment plans	treasury instalment plans, extended for sale contracts
Payroll, casual workers, attendance	erp-api payroll and casual payment modules
SMS, WhatsApp and email with delivery status	notifications-api

SECTION 7

Unit sales and instalments

Every unit sale, from reservation to title, is tracked against the unit, with the buyer able to see exactly what has been paid and what is next.

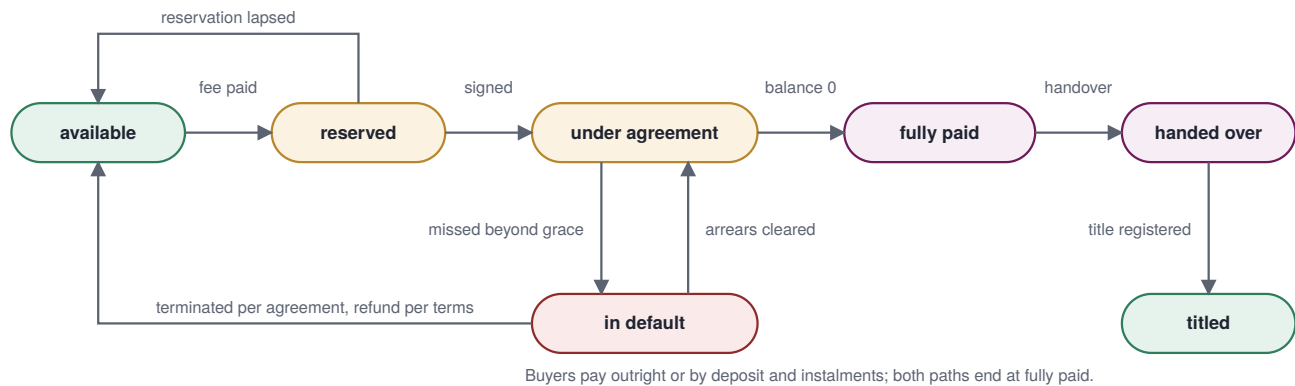


Figure 3. Unit sale lifecycle

7.1 Payment options

Option	How it is recorded
Outright purchase	One invoice for the balance after the reservation fee, due on the agreed date. Part payments are accepted and shown on the statement.
Deposit and monthly instalments	Deposit invoice, then a schedule of equal instalments over the agreed term. The last instalment absorbs rounding.
Milestone instalments	For units bought before completion: instalments fall due on agreed construction milestones, each released by the sales officer with evidence.
Mortgage or SACCO financing	Deposit from the buyer; the financier's payment recorded on release, referenced to the financier's letter of offer.

7.2 Worked example

Item	KES
Three bedroom apartment, price	7,500,000
Reservation fee, credited to deposit	100,000
Deposit at 20%, including the reservation fee	1,500,000
Balance over 24 monthly instalments	250,000 each
Total	7,500,000

Figures are illustrative. Prices, deposits, terms and any interest are set by Shaba Village in each sale agreement.

7.3 Rules

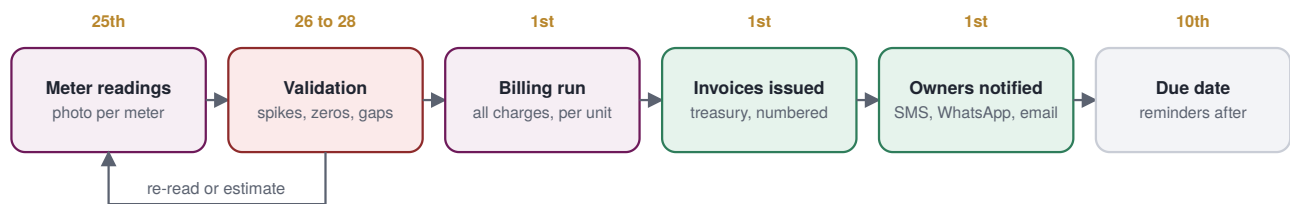
Situation	Rule
Payment received	Applied to the oldest due instalment, then the next. An overpayment reduces the following instalment unless the buyer asks for early completion.
Missed instalment	Reminders at 3 days before, on the due date, and 7 and 14 days after. A late charge applies only if the agreement provides one, at the agreed rate, never compounded.
Default	After the grace period in the agreement, the contract moves to default and the sales officer is alerted with the full history. Notices and any termination are issued by the advocate under the agreement; the platform records each step.
Restructuring	A new schedule may be agreed. It needs approval, keeps the original schedule on record and is shown to the buyer for acceptance.
Termination and refund	Refund amounts follow the agreement and are paid through treasury under approval. The unit returns to available only after the refund is recorded.
Fully paid	A completion statement and receipt are issued, and the handover checklist opens.
Handover	Meter readings, keys, snag list and signatures recorded. Estate billing starts from that date, prorated for the first month.
Title	Each title stage is logged with date and document until the title is released to the owner.

Buyer identity documents and KRA PINs are needed for the sale agreement and title transfer. They are encrypted, visible only to the sales role, and kept for the statutory retention period.

SECTION 8

Billing, utilities and collections

8.1 Monthly cycle



A billing run is keyed by estate and month, so a retry never issues a second invoice for the same unit and period.

Figure 4. Monthly billing cycle

Dates are configurable. The billing run asks treasury to create one invoice per unit and fund with the month's line items. Each invoice carries the unit code as its payment reference and a link to pay. Owners receive a short message with the amount, due date and paybill details, and the full bill in the portal.

8.2 Meter readings

1. The caretaker opens the reading round on a phone. Units are listed in walking order by block.
2. Each reading is typed and photographed. The photo is stored with the reading and shown on the bill.
3. The platform flags readings lower than the previous one, zero consumption for an occupied unit, and consumption above three times the unit's 3-month average.
4. Flagged readings are re-checked. A meter that cannot be read is billed at the 3-month average, marked estimated, and corrected on the next actual reading.
5. A meter replacement records the closing reading of the old meter and the opening reading of the new one, so no consumption is lost or billed twice.

8.3 Water balance

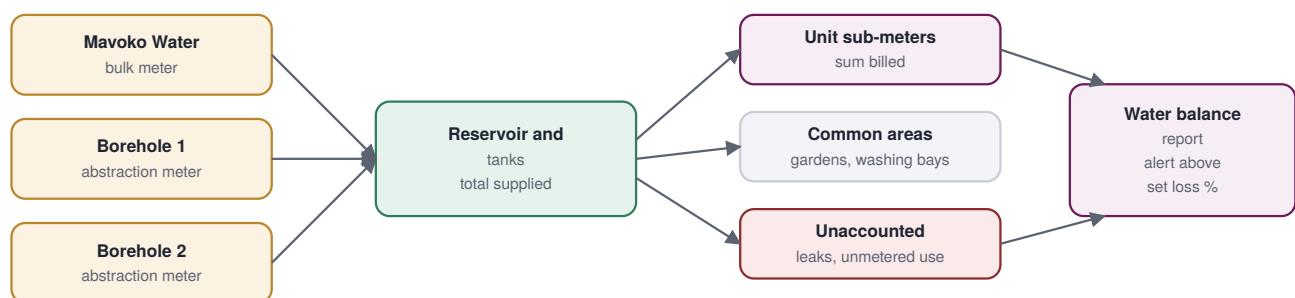


Figure 5. Water balance

Bulk readings from the Mavoko connection and each borehole meter are entered monthly. The report compares total water supplied with water billed to units and estimated common area use. The difference is unaccounted water, shown as a percentage and trended by month. Above the set threshold, for example 15%, the estate manager is alerted to check for leaks.

8.4 Payment matching and allocation

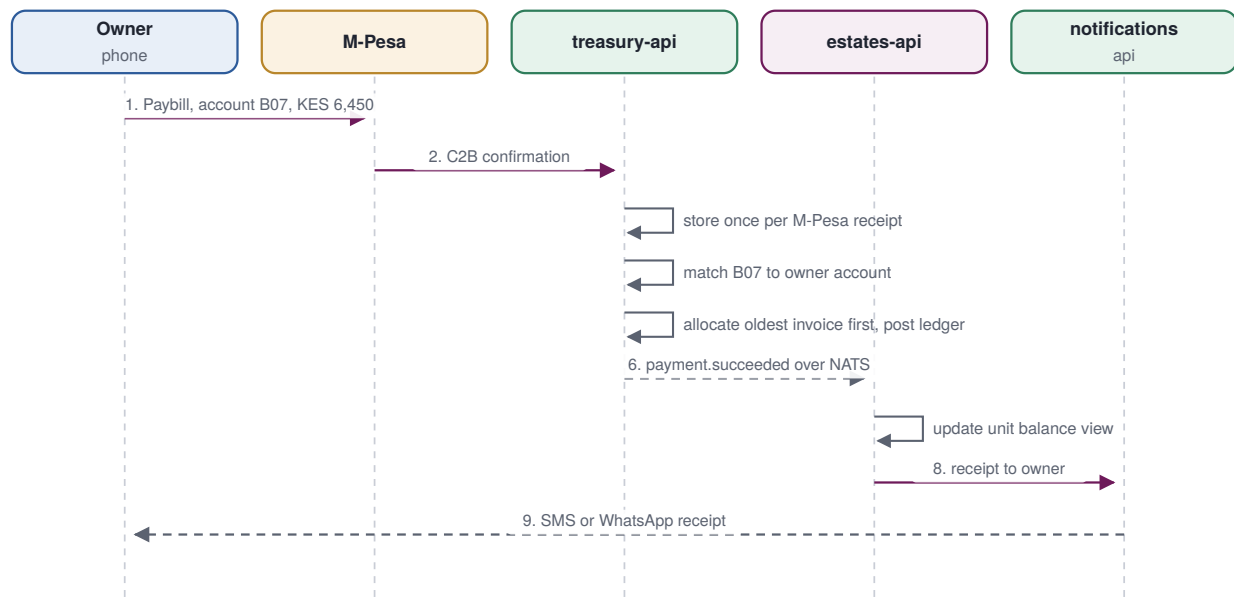


Figure 6. Paybill payment sequence

- Each M-Pesa confirmation is stored once under its receipt number, so a repeated callback never creates a second payment.
- The account number is normalised (case, spaces, dashes) and matched to the unit. Close misses, such as b 07 or B7, are matched; anything else goes to the suspense queue.
- Allocation follows the configured order: oldest invoice first by default, or charge type priority if the by-laws require it. Late charges are settled last.
- Daily, treasury compares the paybill statement with recorded payments. Differences are listed for finance.

8.5 Arrears

Days after due date	Action
1	Friendly reminder with balance and pay link AUTO
7	Second reminder AUTO
14	Firm reminder; late charge applied if the by-laws provide AUTO
30	Call list for the finance officer; offer of a payment plan
45	Demand letter generated for review and sending
60 and above	Escalation to management; any measures permitted by the by-laws, such as withholding a clearance certificate

Not automated: disconnection of water or any other supply, locking of gates or denial of entry to a resident for arrears. These require due process and are excluded from the platform by design.

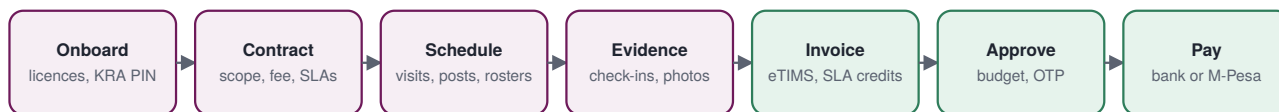
8.6 Budget and transparency

The annual service charge budget is entered in treasury by cost line: security, cleaning, garbage, landscaping, water production, common power, repairs, management, insurance and the sinking fund. Each vendor bill and payroll cost is posted to the matching line. Owners can see a monthly summary of budget against actual, and the income and expenditure statement is produced for the AGM directly from the ledger.

SECTION 9

Service providers and estate operations

Security, garbage, cleaning and maintenance are bought from outside providers. The platform makes each contract measurable and pays only for service delivered.



estates-api runs the first four steps. treasury-api holds the vendor bill, the approval and the payment.

Figure 7. Service provider cycle

9.1 Provider types

Provider	Required documents	Schedule and evidence	Typical SLA
Security agency	PSRA licence, business permit, KRA PIN, insurance	Posts and shifts, roster, guard sign-on, patrol scans, incidents	Every post manned 24 hours; patrol rounds hourly at night; incident report within 1 hour
Garbage collector	NEMA waste transport licence, county permit, KRA PIN	Collection days, check-in photo at each collection point	Collection on scheduled days by 10:00; segregated collection
Cleaning agency	Business permit, KRA PIN, insurance	Daily or weekly rosters by block, checklist with photos	Checklist complete by 09:00; complaints resolved within 24 hours
Landscaping	Business permit, KRA PIN	Weekly visits by zone	Visit completed in its week
Borehole and pump service	Business permit, KRA PIN; WRA driller registration for borehole works	Quarterly service, call-outs as work orders	Breakdown response within 4 hours
Plumbing, electrical, fence, gate	NCA or EPRA licence as applicable, KRA PIN	Work orders	By work order priority
Pest control	PCPB licence, KRA PIN	Quarterly treatment	Visit completed in its month
Tank and reservoir cleaning, water testing	Business permit, accredited laboratory for tests	Twice yearly cleaning; test results uploaded	Results within limits

9.2 Security agency operations

1. Posts are defined with the number of guards per shift, for example main gate two by day and two by night, and a night patrol.
2. The agency supervisor uploads the weekly roster through the vendor portal. Only guards registered for the estate can be rostered.
3. Guards sign on and off at the gate tablet with their badge and PIN. The platform compares sign-ons with the roster and contract and reports unmanned post time.
4. Patrol rounds are scheduled; guards scan QR tags at each checkpoint. Missed checkpoints are reported in the morning summary.
5. Incidents and handover notes are written in the digital occurrence book, visible to the estate manager in real time.
6. At month end, the coverage report is attached to the agency's invoice with any service credit due under the contract.

9.3 Agency personnel

The agency is the employer of its guards and cleaners and remains responsible for their vetting, pay and statutory obligations. Shaba Village records only what it needs to control access: name, agency, role, phone number and badge. A photo is stored only if the agency provides it for identification at the gate. When a person leaves the estate deployment, the badge is deactivated the same day.

9.4 Maintenance work orders

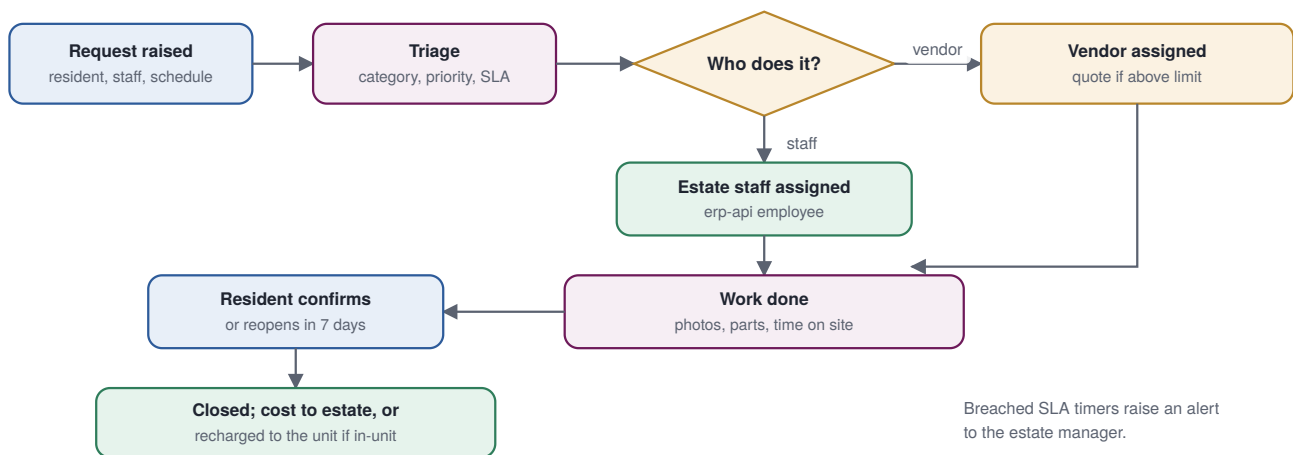


Figure 8. Work order flow

Priority	Examples	Response	Resolution
Emergency	Burst main, sewer overflow, gate or fence down, no water estate-wide	1 hour	4 hours
High	Pump failure, block without water, security light failure	4 hours	24 hours
Normal	Leaking common tap, drainage blockage	24 hours	72 hours
Low	Painting, minor landscaping	3 days	14 days

Repairs inside a unit are the owner's responsibility under most by-laws. Where the estate carries one out at the owner's request, or to stop damage to other units, the cost is recharged to the unit with the work order as evidence.

9.5 Vendor invoices and payment

1. The vendor submits the monthly invoice in the vendor portal with its eTIMS invoice number and document.
2. The platform places the invoice beside the contract fee, completed and missed visits, post coverage and resulting service credit.
3. The estate manager confirms the service; treasury records the vendor bill against the budget line.
4. Approval follows the payout policy: below the limit it is approved by finance, above it by a director with OTP, bound to amount and payee.
5. Treasury pays from the estate account by bank transfer or M-Pesa, deducts withholding tax where it applies, and sends the remittance advice.

9.6 Estate staff

Shaba Village's own regular and casual staff, such as the caretaker, plumber, gardeners and day labourers, are managed in the ERP. Regular staff are paid through monthly payroll with PAYE, NSSF, SHIF and the affordable housing levy. Casual workers are paid per day or per task from the casual payments module, by M-Pesa, with the payment recorded against the work order's cost line. Payroll and casual payments post to the estate fund ledger in treasury. estates-api only stores the ERP employee ID on work orders and service checks.

SECTION 10

Security, access and visitors

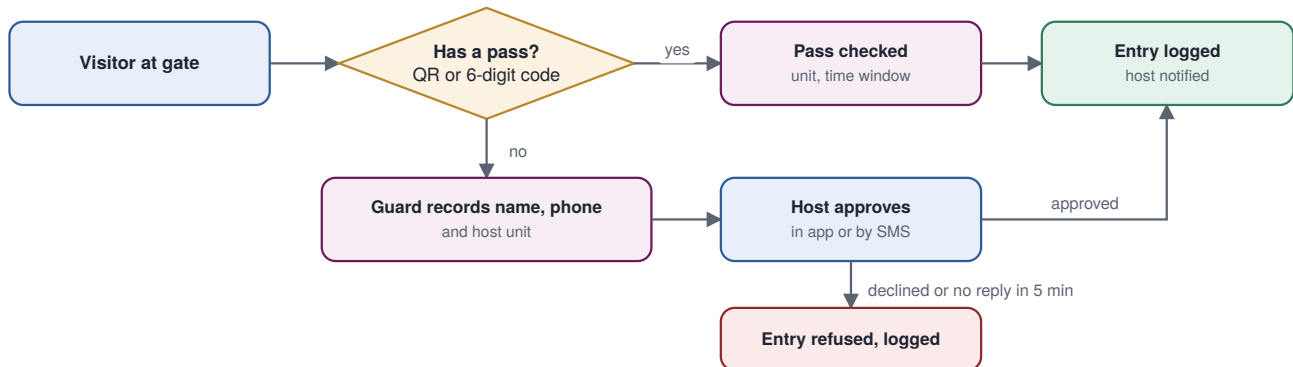


Figure 9. Visitor entry

10.1 Pass types

Pass	Created by	Validity
Guest, single visit	Resident	One entry within the chosen window, default 12 hours
Guest, recurring	Resident	Set days and hours, up to 90 days
Domestic staff	Owner	Set days and hours, renewable, cancellable at any time
Delivery	Resident, or guard with host approval	Single entry, 2 hours
Contractor	Estate manager from a work order	Until the work order closes
Agency personnel	Estate manager	While deployed

10.2 Data collected at the gate

The gate collects the minimum needed for security: the visitor's name, phone number, host unit, time in and out, and vehicle plate if driving. Identity cards are not photographed, copied or retained. Where the by-laws require an identity check, the guard views the card and records only that it was seen. A privacy notice is displayed at each gate and linked in every pass message. Gate events are kept for 90 days and incident records for the period set in the retention schedule.

10.3 Offline operation

Each gate tablet keeps a cache of passes valid for the next 24 hours and of active badges. During a network outage, passes are verified from the cache and new entries are queued on the device, encrypted, and sent when the connection returns. A tablet offline for more than 15 minutes raises an alert to the estate manager.

10.4 Incidents and emergencies

- Incidents are categorised (intrusion, theft, fire, medical, dispute, damage, other) and timed, with photos.
- Serious incidents notify the estate manager and the agency supervisor at once by WhatsApp and SMS.
- Emergency broadcasts reach all residents in a block or the whole estate in one action.

SECTION 11

Process workflows

Workflows not covered in the preceding chapters. Steps marked **AUTO** run without user or staff action.

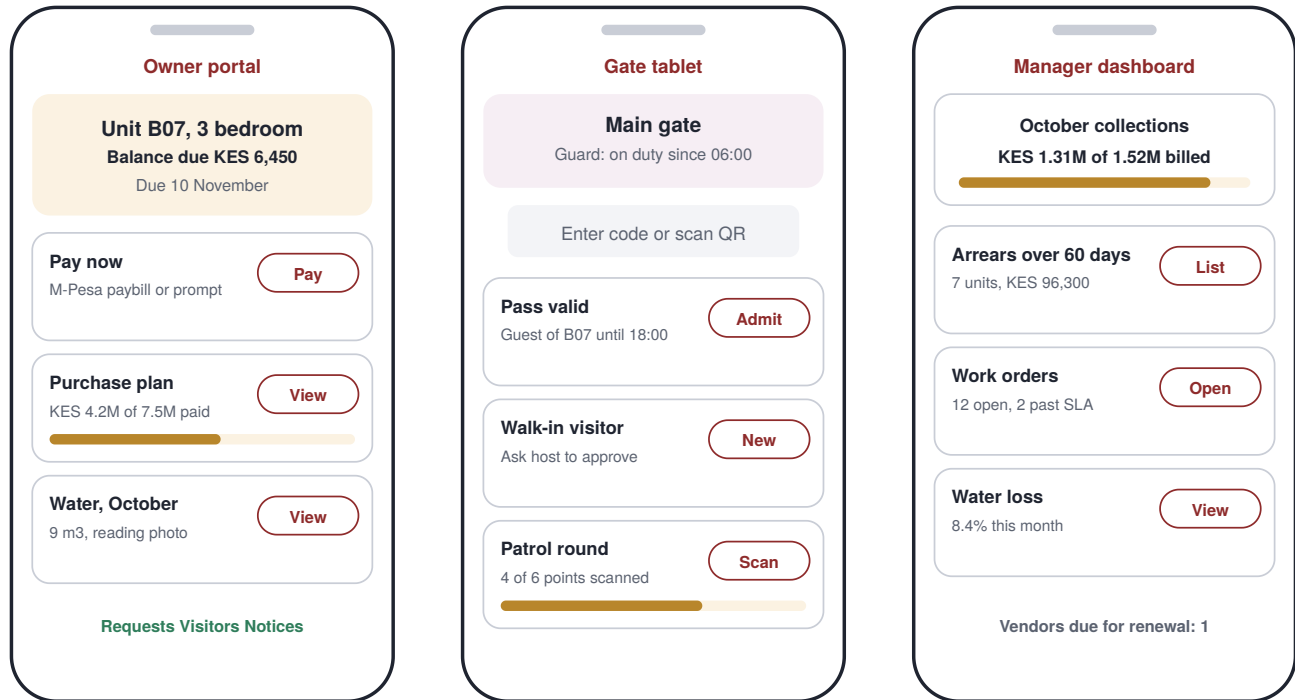


Figure 10. Key screens (low fidelity)

11.1 Estate set-up

Step	Action	Responsible
1	Create the Shaba Village tenant, assign the plan and create staff accounts with roles.	Codevertex
2	Connect the estate and sales paybills (Daraja C2B and prompt) and record both bank accounts. Map each to its ledger and cost centre.	Codevertex, Shaba finance
3	Configure charges, water tariff, penalties, allocation order, reminder days and approval limits.	Shaba finance
4	Import units, owners, occupants, sale contracts and opening balances; reconcile totals to Shaba's records before commit.	Codevertex, Shaba finance
5	Load vendors, contracts, licences, posts, checkpoints and preventive schedules.	Shaba management
6	Load estate staff into the ERP with contracts and pay details.	Shaba management
7	Run one billing cycle in parallel with the current method and compare unit by unit before going live.	Codevertex, Shaba finance
8	Live KES 1 tests on both paybills, and a test vendor payment, each reconciled.	Codevertex, Shaba finance

11.2 Owner onboarding

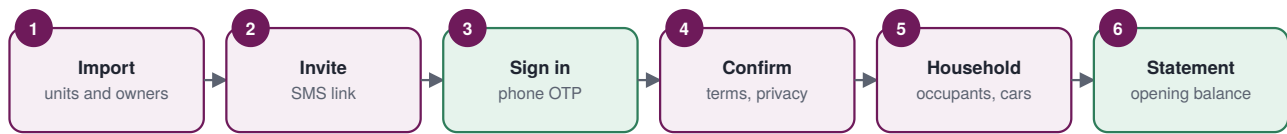


Figure 11. Owner onboarding

Step	Action	System behaviour
1	Owner records imported	Owner linked to unit with ownership date and opening balance
2	Invitation sent	SMS with portal link and paybill details AUTO
3	Owner signs in	One-time password by SMS, WhatsApp fallback; codes hashed, 5-minute expiry, rate limited
4	Owner accepts terms and privacy notice	Version and time recorded
5	Owner adds household members, tenants, vehicles and domestic staff	Gate access created for each
6	Owner views statement	Opening balance and current invoices from treasury

Exception: owners who do not use smartphones continue to receive bills and receipts by SMS and pay by paybill. Staff can create visitor passes on their behalf.

11.3 Unit resale or transfer

1. The selling owner requests a clearance certificate.
2. The platform checks that all estate balances are nil and issues the certificate with a reference and expiry. **AUTO**
3. On the transfer date, the old ownership closes, its account is finalised, and the new owner is invited. Meter readings are taken on the day so water is split correctly.
4. Service charge for the month is apportioned by days between the two owners.

11.4 Owner letting to a tenant

1. The owner adds the tenant as an occupant with start and end dates.
2. The owner chooses which charges the tenant receives (water, garbage, parking).
3. The tenant signs in by phone, creates visitor passes, raises requests and pays assigned bills with the unit account number.
4. On the end date, the tenant's access ends automatically. A final meter reading is requested. **AUTO**

11.5 Notices and alerts

1. The estate manager writes a notice, selects the audience (estate, block, owners, occupants) and channels.
2. Emergency alerts are sent immediately on all channels. Routine notices respect quiet hours, 21:00 to 07:00.
3. Delivery status per recipient is shown; failed SMS falls back to WhatsApp where available.

11.6 Annual budget and AGM

1. Finance prepares the next year's budget by cost line in treasury, with the prior year's actuals beside it.
2. The proposed service charge per unit type is calculated from the budget and unit entitlements.
3. The AGM pack (notice, budget, income and expenditure statement, sinking fund statement) is published in the document library and notified to owners.
4. Rates approved at the AGM are entered with their effective date and apply from the next billing run.

11.7 Handover to the management corporation

1. The estate fund ledger, unit accounts, sinking fund balance, vendor contracts, asset register and documents are exported.
2. The corporation is set up as its own tenant, or Shaba Village continues as managing agent under a new mandate.
3. Paybills and bank accounts are transferred or mandates changed, and opening balances are carried over and reconciled.

11.8 Data subject requests

1. An owner, occupant, visitor or agency worker submits a request for access, correction or deletion.
2. Identity is confirmed by one-time password and the statutory deadline is recorded.
3. Access requests produce a machine-readable export.
4. Deletion erases or anonymises personal data. Financial, sale and tax records required by law are retained for the statutory period, and the requester is told what is kept and why.

SECTION 12

Data design

estates-api owns the estate domain and references other services by ID: users in auth-api, accounts, invoices, plans and vendors in treasury-api, and employees in erp-api.

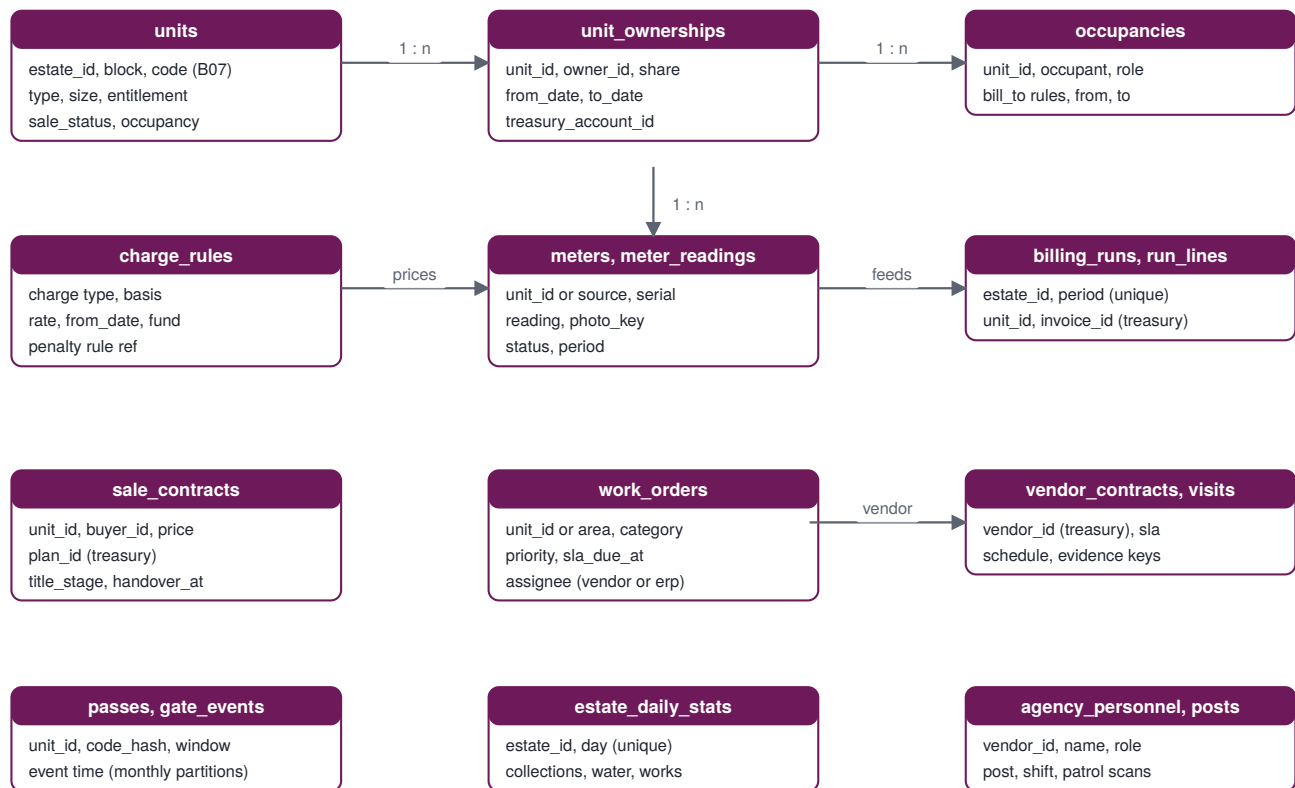


Figure 12. estates-api data model

12.1 Data owned elsewhere

Data	Owner	Held in estates-api
Login identity, sessions, roles	auth-api	User ID
Invoices, payments, balances, statements, instalment plans, vendor bills, ledger	treasury-api	Account, invoice, plan and vendor IDs
Staff records, attendance, payroll, casual payments, assets	erp-api	Employee and asset IDs
Messages and delivery status	notifications-api	Message ID where audit requires

12.2 Scalability and performance measures

Measure	Purpose
Tenant and estate ID with query-specific composite indexes on every table, versioned Atlas migrations	Consistent query speed as data grows; controlled schema change

Keyset pagination on all lists	Constant response time regardless of page depth
Gate events and patrol scans in monthly partitions, purged by partition at the end of retention	Fast writes and cheap deletion for the fastest-growing tables
Unique keys on (estate, period) for billing runs, (unit, period) for readings and run lines, and (event ID) for consumed events	No duplicates from retries or redelivered events
Daily aggregate table per estate, updated by events and rebuilt nightly	Dashboards and reports independent of raw data volume
Read replica for reports and exports; dashboard results cached for 60 seconds and invalidated by events	Reporting never slows the gate or payments
Bulk operations (billing, reminders, imports) in batches of 500 with progress tracking	Predictable memory and time at any estate size
Identity numbers and KRA PINs encrypted at field level; phone numbers hashed for matching	Data protection
Photos and documents in object storage with signed, short-lived links; keys only in the database	Small, fast database; private files

SECTION 13

Interfaces, events and notifications

13.1 estates-api endpoints

Area	Endpoints	Access
Register	GET, POST /units; GET, PATCH /units/{id}; /units/{id}/ownerships, /occupancies, /vehicles; POST /imports	Staff; owners for own unit
Charges and billing	GET, PUT /charge-rules; POST /billing-runs; GET /billing-runs/{id}; POST /adjustments	Finance
Meters	GET /reading-rounds/{period}; POST /meters/{id}/readings; GET /water-balance	Caretaker, finance
Sales	GET, POST /sale-contracts; POST /sale-contracts/{id}/schedule, /restructure, /handover, /title-stages	Sales; buyer read-only
Owner portal	GET /me/units, /me/statement, /me/purchase; POST /me/passes, /me/requests, /me/queries	Owner, occupant
Works	GET, POST /work-orders; PATCH /work-orders/{id}; POST /work-orders/{id}/complete, /confirm, /reopen	Staff, vendor, resident
Vendors	/vendors/{id}/contracts, /documents, /visits, /personnel, /rosters, /invoices	Staff; vendor for own records
Gate	POST /gate/verify, /gate/events, /gate/sync; /patrols/scans; /incidents	Registered gate device
Communication	POST /notices; GET /documents	Staff; residents read
Reports and privacy	GET /reports/{name}; POST /privacy/export, /privacy/delete	By role

13.2 Treasury interfaces used

Function	Treasury capability
Owner and buyer accounts	Customer accounts and balances, one per unit and fund
Monthly and one-off invoices, credit notes	Invoice creation (service to service), recurring schedules, credit notes
Payments	Payment intents with M-Pesa prompt; paybill C2B confirmations matched by account number
Instalments	Instalment plans linked to the sale contract
Reminders	Dunning configuration and reminder events
Vendors	Vendor records, vendor bills, approvals, payouts, withholding tax
Budgets and statements	Budget lines, owner statements, income and expenditure, trial balance

13.3 Events

Event	Publisher	Consumer action
treasury.payment.succeeded	treasury	estates-api refreshes the unit balance view, updates purchase progress and requests a receipt
treasury.invoice.created	treasury	estates-api marks the run line as issued; notifications sends the bill
treasury.dunning.reminder_sent	treasury	notifications delivers the reminder; estates-api records it on the unit timeline
treasury.payout.completed	treasury	estates-api marks the vendor invoice as paid and notifies the vendor
estates.billing_run.completed	estates-api	Finance summary; dashboard refresh
estates.work_order.created, assigned, completed	estates-api	Notifications to resident, assignee and manager
estates.visitor.arrived	estates-api	Host notification
estates.incident.reported	estates-api	Alert to manager and agency supervisor
estates.vendor.document_expiring	estates-api	Alert to manager and vendor
estates.unit.handed_over	estates-api	Estate billing starts; welcome message to the owner

Consumers are idempotent on event ID. Events are held in the outbox during broker outages and delivered on recovery.

13.4 Notifications

Message	Recipient	Channel
One-time password	Owner, occupant, vendor	SMS; WhatsApp fallback
Monthly bill, reminders, receipts	Bill-to party	SMS or WhatsApp; email copy
Instalment due, received, missed	Buyer	SMS or WhatsApp; email copy
Visitor pass and arrival	Visitor; host	SMS or WhatsApp
Work order updates	Requester, assignee	WhatsApp or SMS
Notices and emergency alerts	Selected residents	SMS, WhatsApp, email, in-app
Licence expiry, missed visits, SLA breaches, incidents	Estate manager, vendor	WhatsApp, email
Approval requests, daily finance summary	Finance, directors	Email, WhatsApp

SECTION 14

Reporting and analytics

Report	Audience	Content
COLLECTIONS		
Collections summary	Management, finance	Billed, collected and collection rate by month, charge type and block
Arrears ageing	Finance	Balance per unit in 0 to 30, 31 to 60, 61 to 90 and over 90 day bands, with last payment date
Payment channel report	Finance	Paybill, prompt and bank receipts; unmatched and suspense items
Owner statement	Owner, finance	Invoices, payments and balance per fund
SALES		
Sales position	Directors, sales	Units by status and type; sold value; collected and outstanding
Instalment receivables	Directors, finance	Expected receipts by month for the next 24 months; overdue instalments; buyers in default
Purchase statement	Buyer, sales	Price, payments with receipts, balance, schedule
Handover and title tracker	Sales	Units awaiting handover, snag items open, title stage per unit
UTILITIES AND OPERATIONS		
Water consumption and balance	Management	Consumption by unit and block, estimated readings, unaccounted water percentage trend
Maintenance performance	Management	Work orders by category, time to respond and resolve, SLA breaches, cost by asset
Vendor scorecard	Management	Visits delivered and missed, post coverage, incidents, complaints, service credits
Security summary	Management	Visitors by day and hour, incidents by type, patrol completion
FINANCE		
Budget against actual	Management, owners (summary)	Each cost line, monthly and year to date
Income and expenditure, sinking fund statement	AGM	From the estate fund ledger
Trial balance, cash book, bank reconciliation	Finance, auditors	treasury financial reports for each fund

Reports run on aggregate tables and the read replica, never on the live transaction path. Each figure on a dashboard links to the records behind it, and totals reconcile to the ledger because invoices and payments are read from treasury rather than recalculated.

SECTION 15

Security

Threat	Controls
Account takeover	OTP rate limits by number and IP, hashed codes, short expiry, session revocation, MFA for staff, alerts on new device sign-in for finance roles
One resident seeing another's data	Every request checked against the caller's unit links on the server; IDs alone never grant access; tests for each role
Payment misallocation or fraud by staff	Adjustments and credit notes above a limit need a second approver; every allocation change logged; daily paybill reconciliation
Vendor payment redirection	Payout details verified and locked; changes need MFA and director approval; approvals bound to amount and payee
Duplicate invoices or payments	Unique run keys, receipt-number deduplication, idempotency keys on every write
Gate misuse	Tablets registered to the estate, guard PIN per shift, pass codes hashed and time-bound, one-time passes consumed on entry
Injection and cross-site scripting	Parameterised queries, output encoding, Content Security Policy, sanitised input and file type checks on uploads
Automated abuse	Rate limiting, STK prompt guard, Cloudflare WAF
Data exposure	TLS, encryption at rest, field encryption for identity numbers, least-privilege roles, signed short-lived file links, staff access logging

15.1 Production access

Control	Implementation
Who has access	A named list of Codevertex platform engineers, provided to Shaba Village and updated within 5 working days of any change. No shared accounts.
Server access	Key-based SSH only, behind a default-deny firewall
Changes to production	Only through version-controlled GitOps deployment (ArgoCD); every change traceable to a reviewed commit
Secrets	Sealed secrets; separate database credentials per service; Daraja credentials encrypted in the database
Network isolation	Kubernetes network policies around payment, authentication and estate services
Shaba Village staff	Role-based access through single sign-on with MFA; approvals require a one-time password
Audit	Logins, approvals, settings and financial changes in tamper-evident audit logs, available to Shaba Village

15.2 Codevertex access to estate data

Codevertex staff do not access Shaba Village's personal or financial records in normal operation. Access occurs only to resolve a support request, investigate an incident or perform a restore, is limited to the records concerned, and is

logged. These obligations are set out in the data processing agreement, and every engineer with production access signs a confidentiality undertaking.

15.3 Security incident notification

Event	Timeline
Codevertex notifies Shaba Village of a confirmed personal data breach	Within 24 hours of confirmation
Shaba Village notifies the Data Protection Commissioner (section 43, Data Protection Act)	Within 72 hours of becoming aware; Codevertex supplies the facts
Affected people notified where there is a real risk of harm	Without undue delay, with Shaba Village's approval of the message
Written incident report: cause, impact, remediation	Within 5 working days

15.4 Penetration testing

An internal review against the OWASP Application Security Verification Standard is included in week 13. An independent penetration test is recommended before launch and annually, commissioned at Shaba Village's cost; Codevertex remediates all high and critical findings at no charge before launch.

15.5 Data processors

Processor	Purpose	Data received	Location
Codevertex Africa Limited	Software, hosting operation, support	All platform data	Kenya (operations); EU (servers)
Contabo GmbH	Server infrastructure	Platform data stored on its servers	Germany (EU)
Cloudflare, Inc.	Edge network, firewall, offsite backups	Traffic in transit; encrypted backup files	Global edge; EU and US
Safaricom PLC	M-Pesa collections and payouts	Payer number and name, amount, account number	Kenya
Africa's Talking	SMS delivery	Recipient number, message text	Kenya
Meta Platforms	WhatsApp messaging	Recipient number, message text	Global

SECTION 16

Hosting, backup and disaster recovery

Where Shaba Village's data is held, how it is backed up and how service is restored after a failure. Hosting is included in the monthly fee.

16.1 Hosting location

Component	Provider and location	Content
Application and database servers	Contabo Cloud VPS, European Union region (Germany), running Kubernetes (k3s)	estates-ui, estates-api, treasury, ERP, authentication, notifications; PostgreSQL databases
Offsite backups	Cloudflare R2 object storage, separate provider; encrypted	Nightly encrypted database backups
Edge network	Cloudflare (TLS, CDN, web application firewall)	Traffic in transit and cached public assets; no database content
Email	Codevertex mail server on the same platform	Outbound email; no third-party email processor
Money	Safaricom and Shaba Village's banks	Funds never reside on Codevertex infrastructure

Hosting outside Kenya is a cross-border transfer under sections 48 to 50 of the Data Protection Act. It is documented in the DPIA with its safeguards: provider security certification, encryption in transit and of backups, access restriction and a data processing agreement. Hosting in a Kenyan data centre is available on request and would be quoted as a change to the monthly fee.

16.2 Backup schedule

Backup	Frequency	Scope	Storage	Retention
Platform recovery backup	Daily, 05:00 EAT	All platform databases, verified after each run	Server disk and Cloudflare R2	Replaced only after the new copy is verified offsite
Shaba Village tenant backup	Daily, 05:00 EAT	Shaba Village's estate, treasury and ERP data, encrypted	Cloudflare R2, with optional mirror to Shaba's own storage (Google Drive, OneDrive, S3, SFTP)	7 daily and 4 weekly copies
Payment records	Continuous	Every M-Pesa collection and payout	Safaricom and the banks, independently of Codevertex	Provider retention

16.3 Recovery objectives

Objective	Target	Meaning for Shaba Village
Recovery point objective (RPO)	24 hours	In the worst case, total loss of the server, data is restored from the last nightly backup. Payments made since then are recovered from M-Pesa statements and re-matched, so no payment is lost. Meter readings, work order updates and gate logs from that window may need to be re-entered or re-synchronised from the gate tablets.

Recovery time objective (RTO)	4 hours	Service is restored on a replacement server within 4 hours of a declared disaster.
Optional enhanced RPO	15 minutes	Continuous database log archiving to offsite storage, quoted on request.

16.4 Restore testing

A restore drill is performed quarterly: the latest offsite backup is restored into an isolated environment, record counts and ledger balances are compared with production, and a short report is shared with Shaba Village. The first drill takes place before launch, in week 13.

16.5 Platform outage response

Scenario	Impact	Response
Service or pod failure	Seconds to minutes	Automatic restart; two replicas per service; health checks with alerting
Server failure or provider outage	Portal unavailable	Rebuild on a replacement server from version-controlled configuration, restore of the latest backup and re-matching of M-Pesa payments
During any outage	Owners cannot sign in	Paybill payments still reach Shaba's account; any confirmation missed during the outage is recovered from the M-Pesa statement and matched on recovery. Gate tablets continue offline with cached passes. Shaba Village is notified within 30 minutes of a confirmed incident and updated every 2 hours.

SECTION 17

Legal and regulatory compliance

The legal obligations that apply to Shaba Village's use of the platform and the controls that address them. This is not legal advice. Items marked **Confirm** need an opinion from Shaba Village's advocate before launch.

17.1 Roles and responsibilities

Party	Role	Obligations
Shaba Village	Developer, seller and estate manager; data controller	Sale agreements, by-laws, charges, collections, vendor contracts, ODPC registration, tax, privacy notices, data requests
Codevertex Africa Limited	Software provider and data processor	Secure development and operation, processing on Shaba's instructions under a data processing agreement, breach notification
Service providers	Independent contractors; controllers of their own staff data	Licences, staff vetting and pay, statutory obligations, confidentiality of resident data they see
Safaricom, banks	Payment providers	Collection and payout of funds under their terms
Africa's Talking, Meta, email host	Sub-processors	Message delivery

17.2 Applicable law

Law	Obligation	Control
Constitution of Kenya, Article 31; Data Protection Act, 2019; General Regulations, 2021	Lawful, minimal and transparent processing; data subject rights; DPIA for high-risk processing; 72-hour breach notification; transfer safeguards; consent for marketing; parental consent for children's data	Privacy notices, data minimisation, consent records, DPIA, data request workflow, breach procedure, documented hosting, transactional messages only by default
Data Protection (Registration of Data Controllers and Processors) Regulations, 2021	Property managers, landlords and real estate businesses must register with the ODPC	Shaba Village registers before launch; Codevertex is registered as processor
ODPC draft Guidance Note for the Private Security Sector, 2025	Proportionate visitor data, transparency at gates, CCTV and access logs subject to the Act, retention limits	Name, phone, host and plate only; no ID copies; gate privacy notice; 90-day gate log retention; access by role
Sectional Properties Act, 2020	Management corporation for common property; service charge by unit entitlement; sinking fund; accounts and records; by-laws; AGMs	Entitlement-based charges, separate sinking fund ledger, estate fund books ready for handover, AGM pack, document library
Land Registration Act, 2012; Land Act, 2012; Law of Contract Act	Contracts for disposition of land in writing and signed; registration of transfers and charges	Signed agreements stored; conveyancing by advocates; title stages tracked, not performed

Consumer Protection Act, 2012	Fair terms; clear disclosure of charges; no misleading representations	Charge schedule and penalties published to owners; bills itemised with evidence; query window on every bill
Landlord and tenant law and case law of the Business Premises Rent Tribunal and courts	No self-help disconnection of water or electricity or interference with possession without due process	No automated disconnection or lock-out; arrears handled by reminders, plans and legal process
Water Act, 2016; WASREB; WRA	Abstraction permits for boreholes; water supply in a licensed provider's area; water quality	Permit and water test documents tracked with expiry; bulk and abstraction readings recorded. Confirm the basis on which Shaba supplies and bills water within MAVWASCO's area, and whether a bulk supply agreement is needed.
Energy Act, 2019; EPRA	Only licensed persons sell electricity	Units on Kenya Power prepaid meters; the estate does not resell power; common area power is an estate expense
Sustainable Waste Management Act, 2022	Segregation at source; collection only by NEMA-licensed providers	Collector licence held on file with expiry alert; segregated collection in the SLA
Private Security Regulation Act, 2016 and Regulations, 2019	Security services only by PSRA-licensed firms and registered officers	PSRA licence verified and tracked; only registered agency guards rostered
Estate Agents Act	Persons acting as agents for property sales or letting for others must be registered with the Estate Agents Registration Board	Confirm whether any Shaba staff act as agents for third parties, for example on owner resales or lettings
Proceeds of Crime and Anti-Money Laundering Act, 2009, as amended	Real estate businesses carry customer due diligence, record keeping and reporting duties	Buyer KYC on each sale contract, source of funds field, records kept 7 years, cash payments recorded. Confirm Shaba's reporting obligations and thresholds.
Employment Act, 2007; Occupational Safety and Health Act, 2007; Work Injury Benefits Act, 2007	Contracts, pay records and statutory deductions for estate staff; safety of contractors on site	ERP payroll with statutory deductions; contractor insurance certificates tracked
Income Tax Act; VAT Act, 2013; Tax Procedures Act, 2015; eTIMS	Withholding tax on qualifying payments; eTIMS invoices for deductible expenses since 2026; records for at least 5 years	WHT computed on vendor payments; eTIMS number required on vendor invoices; 7-year retention. Confirm VAT treatment of service charge; residential letting and sale are exempt, and a July 2026 VAT Tribunal decision held owners' service charge contributions not taxable.
Kenya Information and Communications Act; Consumer Protection Regulations, 2010	No unsolicited marketing messages; opt-out	Estate messages are service messages; any marketing needs opt-in with STOP
Computer Misuse and Cybercrimes Act, 2018	Protection of systems and data	Security controls, audit logging, incident response
Children Act, 2022; Data Protection Act, section 33	Children's data only with parental consent and in their best interests	Household members under 18 added only by the owner, name only, for gate access

Persons with Disabilities Act; Constitution, Article 54	Accessible services	WCAG 2.2 AA
EU and UK GDPR (owners in the diaspora)	Equivalent privacy rights where offering services to people in the EU or UK	Kenyan controls applied to all owners; prior consent for optional cookies; documented transfers

17.3 Compliance checklist

#	Requirement	Implementation	Owner
1	Privacy notice	Controller, processors, purposes, lawful bases, retention, rights, transfers, contacts. Linked from every page and message; summary at each gate.	Codevertex draft; advocate approval
2	Owner portal terms	Use of the portal, payment references, bill-to assignment, queries, Kenyan law. Versioned acceptance.	Codevertex draft; advocate approval
3	Charge schedule and penalty policy	Published from by-laws and agreements; matches platform settings.	Shaba Village
4	Gate and CCTV notice	Purpose, data collected, retention, contact.	Codevertex draft
5	Cookie consent	Strictly necessary cookies only at launch; equal accept and reject.	Codevertex
6	Data minimisation	No ID copies at the gate; identity documents only for sale contracts, encrypted.	Codevertex
7	Role-based access	Sales, finance, caretaker, guard and vendor roles see only what they need.	Codevertex
8	Vendor confidentiality	Clause in every provider contract covering resident data seen on site.	Shaba Village
9	No dark patterns	No pre-ticked consents; leaving a pass or request is as simple as creating one.	Codevertex
10	Itemised bills with evidence	Every metered charge shows readings and photo.	Codevertex
11	Accessibility	Contrast 4.5:1, keyboard operation, alternative text, automated checks.	Codevertex
12	Business details	Registered name, address and contacts on every bill, receipt and statement.	Codevertex
13	Children's data	Owner-only entry of minors, name only.	Codevertex
14	Retention and deletion	Schedule below enforced by automated purge; data requests tracked to deadline.	Codevertex
15	Registers	Record of processing, processor list, licence register for fonts and images.	Codevertex

17.4 Record retention

Record	Retention	Basis
Invoices, payments, ledger, vendor bills, payroll	7 years	Tax Procedures Act; AML record keeping
Sale contracts, buyer KYC, title records	7 years after completion of transfer	Contract and AML record keeping
Owner and occupant accounts	While linked to a unit; then 24 months, then anonymised	Data minimisation
Meter readings and photos	Readings 7 years; photos 24 months	Billing evidence
Gate events and visitor details	90 days	Security; ODPC guidance
Incident reports	3 years, or longer if under investigation or claim	Security and insurance
Agency personnel records	12 months after deployment ends	Access control
Work orders	7 years	Asset history and cost evidence
Consent and terms acceptance	Account life plus 5 years	Evidence of consent
One-time passwords	5 minutes, hashed	Security
Security and audit logs	12 months	Incident investigation

17.5 Legal deliverables

- Privacy notice, gate notice, cookie policy, owner portal terms, vendor portal terms
- Data processing agreement and sub-processor list
- Data protection impact assessment and record of processing activities
- Breach response and data request procedures
- Vendor confidentiality clause, accessibility statement

SECTION 18

Project delivery

Delivery follows Agile Scrum with weekly demonstrations on the staging environment.

18.1 Methodology

Practice	Approach
Iterations	One-week Sprint 0, then two-week sprints, a hardening week and a launch week
Demonstrations	Weekly 30-minute review with the product owner; decisions logged
Backlog	User stories traced to FR and NFR identifiers, with acceptance criteria
Definition of done	Code reviewed, tests and accessibility checks passed, documented, deployed to staging, accepted
Data migration	Import template issued in week 1; trial imports from week 3; balances reconciled with Shaba before go-live
Parallel run	One billing cycle produced by both the old method and the platform, compared unit by unit
Release management	CI on GitHub Actions; GitOps deployment through ArgoCD
Change control	Equal-size scope substitutions within the fixed price; other changes estimated and approved in writing

18.2 Project team

Role	Allocation
Project lead and business analyst	Part time, weeks 1 to 14
Full-stack engineers (Go, Next.js)	Two, weeks 1 to 14
Product designer	Part time, weeks 1 to 5; reviews thereafter
QA engineer	Part time, weeks 3 to 14
Platform and DevOps engineer	As required

18.3 Implementation schedule

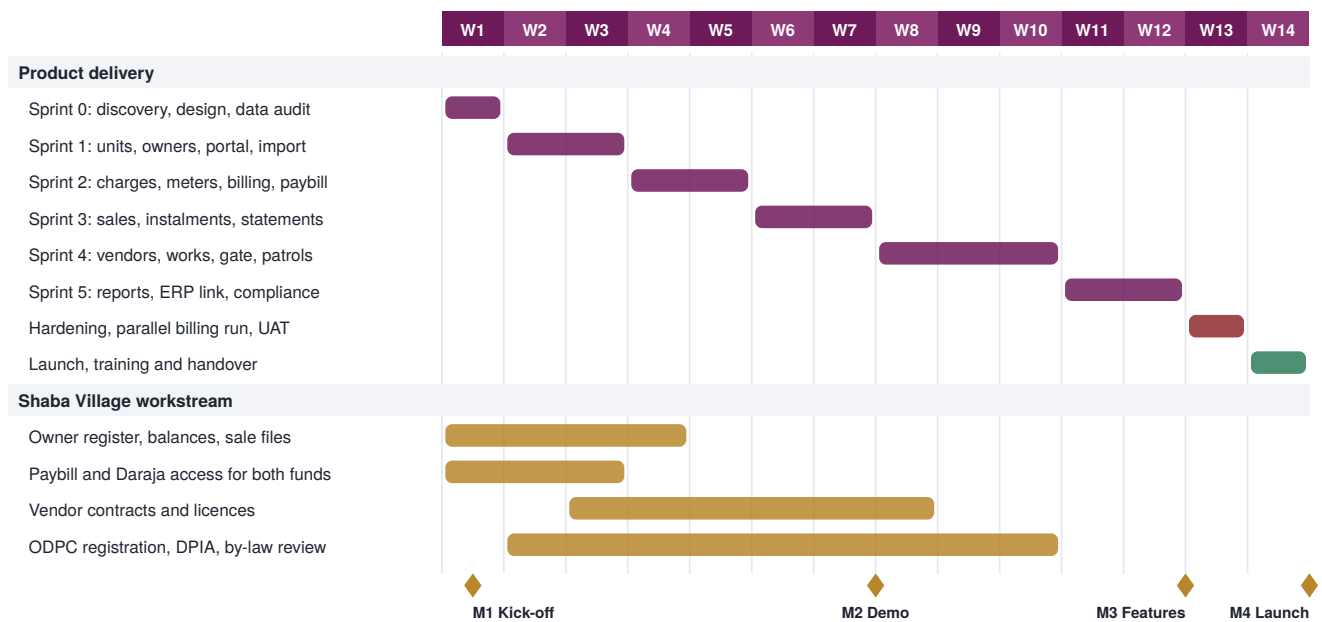


Figure 13. Implementation schedule

Week	Phase	Deliverables
1	Sprint 0	Kick-off, by-law and agreement review, data audit and import template, UX flows, visual design, architecture approval, environments
2 to 3	Sprint 1	Units, owners, occupants, phone sign-in, owner portal shell, imports, staff roles
4 to 5	Sprint 2	Charge rules, meter readings, billing runs, paybill matching and allocation, receipts, statements, reminders
6 to 7	Sprint 3	Price lists, reservations, sale contracts, instalment schedules, purchase statements, handover. Milestone 2: end-to-end demonstration with live paybill payments
8 to 10	Sprint 4	Vendors, contracts, schedules and evidence, vendor portal and invoices, work orders, gate tablet, passes, patrols, incidents
11 to 12	Sprint 5	Reports and dashboards, budgets, ERP staff and casual payments, notices and documents, privacy tools, accessibility. Milestone 3: feature complete
13	Hardening	Security review, load test, restore drill, parallel billing run, UAT, legal pack approval
14	Launch	Production release, owner invitations, training, handover. Milestone 4: launch acceptance

Based on contract signature by 23 October 2026, work begins on 26 October 2026. Weeks 1 to 8 run to 18 December. The team pauses from 21 December to 3 January, and work resumes on 4 January 2027 with launch in the week of 8 February 2027.

SECTION 19

Testing and acceptance

Test	Objective	Exit criteria
Automated tests	Charge calculation, proration, allocation, instalments, SLA credits	All tests pass on every merge
End-to-end tests	Owner, buyer, caretaker, guard, vendor and finance journeys	Pass on Android Chrome, iOS Safari and desktop
Migration reconciliation	Imported balances and contracts	Totals per unit and fund match Shaba's signed-off figures
Parallel billing run	One month billed both ways	Every unit's bill matches, or the difference is explained and accepted
Live payment tests	Both paybills, prompt, vendor payout	KES 1 transactions matched, receipted and reconciled
Offline gate test	Network removed for 2 hours	Passes verified, entries queued and synchronised without loss
Load test	Peak use	Billing run for 1,000 units within NFR-04; 50 requests per second within NFR-02
Security review	OWASP risks, access between units, payment adjustments	No open high or critical findings
Accessibility	WCAG 2.2 AA	Automated and manual checks passed
User acceptance	Business acceptance by Shaba Village staff and pilot owners	All M requirements accepted; no open severity 1 or 2 defects

SECTION 20

Commercial terms

20.1 Fixed price

Work package	KES
Discovery, by-law and data review, UX and visual design, architecture	25,000
Estates service: register, charges, meters, billing runs, sales and instalments, works, vendors, gate and visitors	105,000
Web application: owner portal, management console, gate tablet, vendor portal	80,000
Finance integration: invoices, paybill matching, allocation, instalment plans, statements, reminders, vendor bills, payouts, fund ledgers	55,000
ERP set-up (staff, payroll, casual payments, attendance) and message templates	20,000
Data migration, compliance features, legal pack drafts	30,000
Testing, security review, parallel run, UAT, training, launch	35,000
Total (inclusive of taxes)	350,000

The ERP implementation fee normally charged on a new ERP subscription is included in this price.

20.2 Payment schedule

Milestone	Due	Share	KES
M1 Contract signature	Week 0	30%	105,000
M2 End-to-end demonstration	End of week 7	30%	105,000
M3 Feature complete	End of week 12	30%	105,000
M4 Launch acceptance	End of week 14	10%	35,000
Total		100%	350,000

20.3 Monthly service tiers

The monthly fee follows the Codevertex ERP subscription tiers. Each tier includes the estates platform licence, the ERP and treasury subscription at that tier, hosting, backups, monitoring, security updates, and support and maintenance.

Tier	Units	Estate staff in ERP	Staff users	KES per month
Estate Starter	Up to 150	25	10	10,000
Estate Growth (recommended)	Up to 400	75	30	20,000
Estate Professional	Up to 1,000, several estates or phases	200	100	35,000

Estate Enterprise	Unlimited	Unlimited	Unlimited	By quote
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Owners, occupants, guards and vendor supervisors are not counted as staff users and are unlimited in every tier.

Capability	Starter	Growth	Professional
Register, billing, collections, sales and instalments, owner portal, gate, vendors, work orders, notices	Yes	Yes	Yes
Payroll, casual payments, leave, attendance	Yes	Yes	Yes
Multi-level approvals for vendor bills, credit notes and restructures	Single level	Yes	Yes
Service charge budget against actual, asset register, BI reports	No	Yes	Yes
API access, custom workflows, extended audit trail, priority support	No	No	Yes

Recommendation: Estate Growth at KES 20,000 per month. The budget against actual report (FR-59), multi-level approvals (FR-22, FR-44) and the asset register used by preventive maintenance (FR-46) are Growth features, and they carry most of the transparency owners expect from estate management.

Term	Position
Billing	Monthly in advance from the launch month. Annual prepayment earns one month free.
Changing tier	Upgrades apply at once; downgrades from the next month if usage fits the lower tier.
Usage limits	No per-transaction or per-payment charge from Codevertex.
Price changes	Fixed for 12 months from launch; thereafter at most 10% a year, with 60 days' written notice.

20.4 Client-borne costs

Item	Scope	Rate (KES)
SMS	Bills, receipts, reminders, passes, one-time passwords; prepaid, minimum top-up KES 500	1.00 per SMS
WhatsApp	Bills, receipts, work order updates, alerts; monthly plan	500 to 2,000 per month
M-Pesa charges	Safaricom paybill and payout tariffs on Shaba's own paybills	Safaricom tariff
Domain	Registration and renewal of Shaba Village's portal domain	Registrar rate
Gate tablets and patrol tags	One Android tablet per gate with data; weatherproof QR tags for checkpoints	Supplier rate

20.5 Messaging rates

Plan or item	Allowance	KES
SMS	Per SMS of up to 160 characters	1.00
WhatsApp Basic	100 messages per month	500 per month
WhatsApp Standard	500 messages per month	1,000 per month

WhatsApp Pro	Unlimited messages	2,000 per month
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20.6 Operating cost examples

Monthly item (KES), Estate Growth	120 units	250 units	400 units
Monthly service fee	20,000	20,000	20,000
SMS (5 per unit: bill, reminder, receipt, passes)	600	1,250	2,000
WhatsApp Pro	2,000	2,000	2,000
Total	22,600	23,250	24,000
Cost per unit per month	188	93	60

M-Pesa charges depend on the tariff on Shaba's paybills and are excluded. The cost can be recovered within the management line of the service charge budget.

20.7 Optional services

Service	Scope	Billing
Managed domain	Registration, renewal, DNS and certificates	Annually, at the registrar's rate
Smart water meters	Integration of remote-read meters through the Codevertex IoT service	Quoted on meter choice
Rent collection for owners	Letting management for owners who let units: tenant invoicing, rent receipts and owner remittance	Quoted on request
Kenyan data centre hosting	Hosting within Kenya	Quoted as a change to the monthly fee

SECTION 21

Ownership, licensing and exit

Shaba Village owns its data, documents and money. Codevertex owns the platform, which Shaba Village uses under licence.

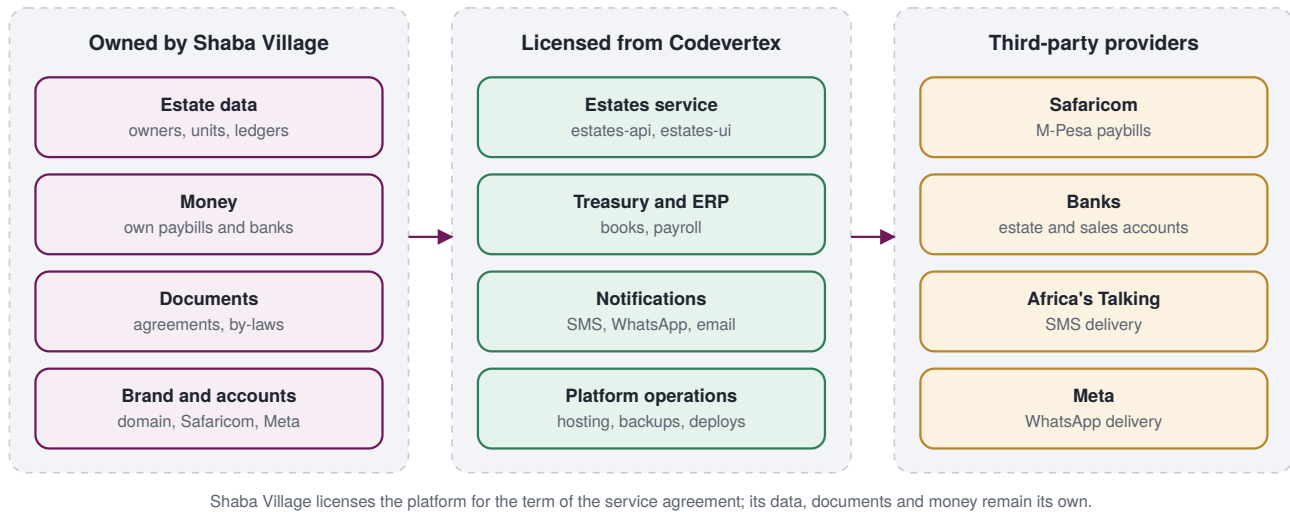


Figure 14. Ownership boundaries

21.1 Ownership

Asset	Owner	Terms
Estate data: units, owners, occupants, readings, invoices, payments, contracts, work orders, gate records, ledgers	Shaba Village	Owned from creation; Codevertex processes it only on Shaba Village's instructions
Shaba Village configuration, templates, documents, designs and content	Shaba Village	Assigned to Shaba Village on payment of the build price
Money	Shaba Village and, for the estate fund, the owners	Held in Shaba's paybills and bank accounts, never by Codevertex
Brand, domain and accounts with Safaricom, banks and Meta	Shaba Village	Registered in Shaba Village's name
Codevertex platform: estates service, authentication, treasury, ERP, notifications, subscriptions, shared libraries, deployment tooling	Codevertex	Proprietary; licensed as below

21.2 Platform licence

Shaba Village receives a non-exclusive, non-transferable licence to use the platform for the duration of the service agreement, included in the monthly fee. Shaba Village's own developers, or another vendor it appoints, may use the documented APIs with keys issued to Shaba Village. The licence passes to the management corporation, at Shaba's direction, on handover of the estate.

21.3 Exit and migration

Step	Commitment
Notice	At least 90 days' written notice before Codevertex discontinues any service Shaba Village relies on.
Data export	Complete export in open formats: database dump of Shaba Village's records, ledgers and statements (CSV and Excel), unit and owner records (JSON), photos and documents.
Money	Unaffected. Funds are in Shaba's own paybills and bank accounts.
Transition assistance	Up to 20 hours of engineering support, included; further support at the standard rate.
Source code escrow (optional)	Source code of the services Shaba Village relies on deposited with an independent escrow agent, released if Codevertex ceases to trade. Agent fees borne by Shaba Village.

21.4 Handover on termination

- Final data export and the latest backups within 10 working days
- Operations documentation, the legal pack and registers
- Transfer of domain and DNS administration, where managed by Codevertex
- Written confirmation of deletion of Shaba Village data from Codevertex systems after export, subject to statutory retention

SECTION 22

Contract terms

Principal terms of the master services agreement between Shaba Village and Codevertex. Codevertex's standard terms apply where not varied below.

Term	Position
Initial term	12 months from launch, renewing monthly thereafter
Termination for convenience	Either party, after the initial term, with 60 days' written notice
Termination for cause	Either party, for a material breach not remedied within 30 days of written notice, or on insolvency
Fee after termination	Payable to the end of the notice period only. Export and handover at no charge.
Service level	99.5% monthly availability for the owner portal, gate service and payment matching, excluding maintenance notified 48 hours ahead
Service credits	Below 99.5%: 10% of the monthly fee. Below 99.0%: 25%. Below 95.0%: 50%.
Prolonged downtime	An outage over 72 continuous hours, or service credits in 2 consecutive months, entitles Shaba Village to terminate without notice and receive full handover.
Delay caused by Codevertex	2% of the build price credited for each full week of delay to launch caused solely by Codevertex, up to 10% (KES 35,000). The final milestone is not due until launch acceptance.
Delay caused by Shaba Village	Where data, decisions, documents, paybill access, feedback or test participation are later than agreed (decisions within 2 working days, documents within 3), the schedule extends day for day with no delay credits. Milestones fall due on their original dates for work completed. Team standby beyond a cumulative 5 working days is charged at KES 25,000 per week, pro rata.
Force majeure and third parties	Events outside both parties' control, including delays by Safaricom or banks after complete documents were submitted on time, extend the schedule day for day with no credits or charges. Beyond 60 days, either party may terminate, and Shaba Village pays for work completed.
Liability cap	Each party's total liability is limited to the fees paid in the 12 months before the claim. Indirect and consequential losses are excluded.
Exclusions from the cap	Breach of confidentiality, the intellectual property indemnity and wilful misconduct
Codevertex indemnity	Against third-party claims that the software infringes intellectual property, and regulatory fines arising from Codevertex's breach of the data processing agreement, within the cap
Shaba Village indemnity	Against claims arising from Shaba Village's charges, by-laws, sale agreements, content and handling of funds
Governing law	Laws of Kenya. Negotiation for 30 days, then mediation, then the courts of Kenya.

SECTION 23

Risks and mitigations

Risk	Impact	Mitigation	Owner
Incomplete or disputed opening balances	Owner complaints at launch	Data audit in week 1, trial imports, reconciliation sign-off, statement review by owners in the first month	Shaba Village, Codevertex
Paybill or Daraja access delayed	Payments not matched automatically	Requested in week 1; manual statement import as fallback	Shaba Village
Water bill disputes	Collection delays	Photo per reading, anomaly checks, query window, meter replacement records	Shaba Village operations
Owners not adopting the portal	Reliance on office queries	SMS bills and paybill work without the portal; launch communication; office assisted sign-up	Shaba Village
Provider resistance to SLA checks	Incomplete evidence	SLAs written into renewed contracts; simple check-in on any phone; supervisor training	Shaba Village
Gate network failure	Slow entry	Offline mode with cached passes; device alerts	Codevertex
Legal position on water supply or agents	Change to billing or roles	Advocate review from week 2; configurable charges	Shaba Village
Scope change	Schedule slippage	Change control; agreed exclusions	Both
Holiday period	Slower decisions	Planned pause; decisions scheduled before 18 December	Both

SECTION 24

Post-launch support

Service	Terms
Hypercare	30 days after launch, covering the first full billing cycle; same-day response to billing and payment incidents
Warranty	Defects against this specification corrected at no cost for 90 days
Ongoing support	Support and maintenance included in the monthly fee: defect correction, configuration help, platform updates
Response targets	Critical (payments, gate, portal down): 1 hour. High: 4 working hours. Normal: next working day.
Monitoring	Alerts on failed billing runs, unmatched payments, offline gate tablets, error rates and latency
Training	Management and finance (two sessions), caretakers, guards and vendor supervisors (one session each), with short video guides
Handover	Administration guide, finance guide, gate guide, runbooks, legal pack, registers
Enhancements	Smart meters, rent collection for owners, owner voting and native applications quoted on request

SECTION 25

Acceptance

Signature below confirms acceptance of the scope, schedule and price in this document. Changes are subject to the change control process in [Project delivery](#).

Shaba Village	Codevertex Africa Limited
Name	
Title	
Signature	
Date	